

## TERM PAPER

# Visualizing Global Patterns of Colorectal Cancer Incidence

## A 2022 Country-Level Analysis with a Middle East Comparative Lens

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### Abstract

**Background:** Colorectal cancer is a major global cancer burden, but its incidence varies strongly across countries and regions. This variation may reflect lifestyle patterns, economic development, screening access, diagnostic capacity, and cancer registry quality. This study examines global country-level colorectal cancer incidence while highlighting the Middle East as a comparative subgroup.

**Methods:** A 2022 cross-sectional ecological dataset of 173 countries was constructed by integrating colorectal cancer incidence age-standardized rates from GLOBOCAN, adult obesity prevalence and adult physical activity prevalence from WHO sources, GDP per capita from the World Bank, and regional labels. The analysis used exploratory visualization, Spearman correlation, Kruskal-Wallis tests, regression modeling, Mann-Whitney U tests, and principal component analysis.

**Results:** Log GDP per capita showed the strongest association with colorectal cancer ASR ( $r_s = 0.677$ ). Obesity had a weaker positive association ( $r_s = 0.281$ ), while physical activity had a weak negative association ( $r_s = -0.177$ ). Regional differences were significant for colorectal cancer ASR, obesity, physical activity, and log GDP ( $p < 0.001$ ). Regression results showed that model fit improved from  $R^2 = 0.036$  with lifestyle variables only to  $R^2 = 0.533$  after adding log GDP and region. The Middle East had higher obesity and lower physical activity than non-Middle Eastern countries, but not significantly higher colorectal cancer ASR.

**Conclusion:** Colorectal cancer incidence in this global dataset aligns more strongly with GDP and regional context than with lifestyle indicators alone. The findings should be interpreted as ecological associations, not causal evidence.

**Index Terms** – colorectal cancer; visual analytics; obesity; physical activity; GDP; ecological study; PCA; Middle East; global health

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# 1 Introduction

Colorectal cancer is one of the major cancer burdens in global public health. Recent GLOBOCAN 2022 estimates show that colorectal cancer remains among the leading cancers worldwide, with more than 1.9 million new cases and substantial variation in incidence between countries and world regions (Bray et al., 2024). Earlier international studies also show that colorectal cancer incidence and mortality vary strongly across countries and development levels (Arnold et al., 2017; Center et al., 2009). This variation is not only a clinical issue. It also reflects differences in lifestyle patterns, economic transition, screening availability, diagnostic access, healthcare-system capacity, and cancer registration quality. For this reason, colorectal cancer is a suitable case for country-level visual analytics, especially when cancer incidence is examined together with lifestyle and economic indicators.

Although many cancer studies focus on highly visible cancer types such as breast cancer, lung cancer, and oral cancer, colorectal cancer is also an important and growing area of research. Its country-level pattern is especially interesting because it sits at the intersection of population lifestyle, health-system development, and detection capacity. In high-income settings, screening and stronger diagnostic systems can increase recorded incidence by detecting more cases. In lower-resource settings, weaker screening coverage, delayed diagnosis, and less complete cancer registration may affect the measured burden. Therefore, international comparisons must be interpreted carefully because recorded incidence can reflect both true disease risk and the ability of a country to detect and report cases.

Lifestyle-related factors are central to colorectal cancer prevention research. Evidence reviews from the World Cancer Research Fund and the American Institute for Cancer Research link higher body fatness with increased colorectal cancer risk and identify physical activity as an important protective factor, especially for colon cancer (World Cancer Research Fund/American Institute for Cancer Research, 2018). Peer-reviewed meta-analyses also support the relationship between obesity and colorectal cancer risk (Larsson & Wolk, 2007; Moghaddam et al., 2007), while physical activity has been repeatedly associated with lower colorectal cancer risk (Shaw et al., 2018). These relationships are often studied using individual-level epidemiological data. In contrast, the present paper uses country-level indicators. Adult obesity prevalence and adult physical activity prevalence therefore represent national population profiles rather than individual exposure histories. This distinction matters because an ecological association between national obesity and national colorectal cancer incidence does not prove individual-level risk.

Economic development is also relevant to the interpretation of colorectal cancer incidence. GDP per capita is not a biological cause of cancer. Instead, it can act as a country-level proxy for two broader processes. The first is *detection capacity*: wealthier countries may have better healthcare access, screening systems, diagnostic infrastructure, and cancer registration. The second is *lifestyle transition*: higher income may also be associated with more sedentary work, greater access to calorie-dense foods, processed foods, and red or processed meat, and other features of urbanized lifestyles. Previous ecological studies have examined colorectal cancer

in relation to development indicators such as the Human Development Index (Goodarzi et al., 2019; Khazaei et al., 2016; Rafiemanesh et al., 2016), and broader global studies also describe colorectal cancer as closely linked with international development and cancer-transition patterns (Arnold et al., 2017; Center et al., 2009). This study uses GDP per capita as a direct economic indicator, with log transformation used to reduce skewness and improve comparison across countries.

The Middle East provides a meaningful comparative lens within this global setting. Several studies have examined colorectal cancer burden, stage at diagnosis, or risk factors in North Africa and the Middle East, the Gulf region, or selected countries (Haghighatdoost et al., 2024; Salimzadeh et al., 2024). A recent scoping review also shows that obesity, physical inactivity, diet, and other modifiable risk factors are important cancer-related concerns in the Middle East and North Africa (Mansour et al., 2024). However, there is still value in positioning Middle Eastern countries within a broader global country-level profile space using the same 2022 cancer, lifestyle, and economic indicators. This paper therefore does not treat the analysis as only a Middle Eastern study. Instead, the global dataset is used as the reference frame, and Middle Eastern countries are highlighted to examine whether they show a distinct or semi-distinct profile compared with the wider international pattern.

This paper adopts a hybrid hypothesis-driven and data-driven approach. The hypothesis-driven component evaluates whether colorectal cancer incidence is associated with adult obesity, adult physical activity, and GDP per capita at the country level. The data-driven component uses exploratory visualization and principal component analysis to identify broader regional and subgroup patterns. This combination is appropriate because the topic is grounded in prior cancer epidemiology, while the integrated country-level dataset also allows exploratory comparison across regions.

The main research question is:

How do adult obesity, adult physical activity, and GDP per capita relate to colorectal cancer incidence across countries in 2022, and how do these patterns differ by world region?

The secondary research question is:

Where do Middle Eastern countries sit within the global colorectal cancer, lifestyle, and economic profile space?

The study tests the following hypotheses:

- **H1:** Countries with higher adult obesity prevalence will tend to have higher colorectal cancer incidence ASR.
- **H2:** Countries with lower adult physical activity prevalence may tend to have higher colorectal cancer incidence ASR.

- **H3:** GDP per capita will be associated with colorectal cancer ASR because GDP may reflect two country-level pathways: detection capacity and lifestyle transition.
- **H4:** Colorectal cancer, obesity, physical activity, and GDP patterns will differ across world regions.
- **H5:** Middle Eastern countries will show a recognizable profile within the global country-level space rather than being randomly distributed across the colorectal cancer, lifestyle, and economic indicators.

The main contribution of this study is a reproducible global visual analytics pipeline that integrates GLOBOCAN, WHO, and World Bank data to examine country-level colorectal cancer incidence patterns, compare regional profiles, and position Middle Eastern countries within the global cancer-lifestyle-economic landscape.

## 2 Data and Sampling Strategy

### 2.1 Study Design

This study uses a 2022 cross-sectional ecological country-level design. Each row in the analytical dataset represents one country. The aim is to examine how colorectal cancer incidence varies across countries in relation to national lifestyle, economic, and regional indicators.

The year 2022 was selected because it was the most recent year that could be aligned across the main data sources: GLOBOCAN 2022, WHO adult obesity estimates, WHO adult physical activity estimates, and World Bank GDP per capita. Using one common year kept the analysis focused and avoided adding a time-series dimension, which would require extra decisions about lag periods, repeated country observations, and changing country coverage.

Because the analysis is ecological, all findings refer to country-level associations. They should not be interpreted as individual-level risk relationships. This is important because a country with high obesity and high colorectal cancer incidence does not prove that individuals with obesity are the same individuals developing colorectal cancer.

### 2.2 Data Sources

The dataset integrates four main international sources. Colorectal cancer incidence was obtained from GLOBOCAN 2022 through the Global Cancer Observatory Cancer Today data tables (Bray et al., 2024; International Agency for Research on Cancer, 2024b). Adult obesity prevalence was obtained from the WHO Global Health Observatory indicator for obesity among adults aged 18 years and older (World Health Organization, 2024b). Adult physical activity prevalence was derived from the WHO Global Health Observatory indicator for insufficient physical activity among adults aged 18 years and older (World Health Organization,

2024a). GDP per capita in current US dollars was obtained from the World Bank World Development Indicators (World Bank, 2024).

The WHO physical activity source variable was originally reported as insufficient physical activity. During preprocessing, it was converted into adult physical activity prevalence so that higher values consistently mean more physical activity:

$$\text{activity\_value} = 100 - \text{inactivity\_value}$$

The final paper therefore refers to adult physical activity prevalence, except when explaining this transformation.

## 2.3 Variables

The main outcome variable is colorectal cancer incidence age-standardized rate, World standard population, per 100,000. In the dataset, this is stored as `asr_world`. A duplicate analytical variable, `cancer_index`, was also created and is equal to `asr_world`.

Age-standardized rates were used instead of crude rates because countries have different age structures. This is especially important for colorectal cancer, since incidence increases strongly with age. Age standardization makes international comparison more meaningful by reducing the effect of population age differences (International Agency for Research on Cancer, 2024a). Since GLOBOCAN ASR already adjusts for age structure, the cancer outcome did not need to be restricted to adults aged 18 years and older. The lifestyle variables were adult-based because obesity and physical activity are adult behavioral and metabolic indicators, and colorectal cancer mainly affects adults and older populations (American Cancer Society, 2023; Centers for Disease Control and Prevention, 2025).

The main explanatory variables were adult obesity prevalence, adult physical activity prevalence, and GDP per capita. Obesity was included because higher body fatness is a recognized risk factor for colorectal cancer (Larsson & Wolk, 2007; Moghaddam et al., 2007; World Cancer Research Fund/American Institute for Cancer Research, 2018). Physical activity was included because it is commonly discussed as a protective factor, especially for colon cancer (Shaw et al., 2018; World Cancer Research Fund/American Institute for Cancer Research, 2018). GDP per capita was included as a country-level economic indicator, not as a biological cause of cancer. It is interpreted as a proxy for two broad processes: *detection capacity*, including healthcare access, screening, diagnosis, and registry quality, and *lifestyle transition*, including more sedentary work and greater access to calorie-dense or processed foods.

GDP per capita was highly right-skewed, so the natural logarithm was used to create `log_gdp`. Region labels were also created to compare broad geographic patterns. The final region categories were Africa, Americas, Asia, Europe, Middle East, and Oceania.

## 2.4 Population, Sampling, and Missing Data

The target population was all countries with available 2022 data for colorectal cancer incidence, adult obesity prevalence, adult physical activity prevalence, GDP per capita, and region classification. The study used a complete-case country-level sample rather than a random sample.

The cancer dataset was used as the base dataset because colorectal cancer incidence is the main outcome. Obesity, physical activity, and GDP data were then merged onto the cancer country list using ISO3 country codes. The initial merged dataset contained 186 country or territory rows. After complete-case filtering on the core analytical variables, the final analytical dataset contained 173 countries. A total of 13 rows were removed because at least one required variable was missing.

## 2.5 Middle East Subgroup Definition

The Middle East subgroup was defined using ISO3 country codes. The countries included in the final analytical dataset were Bahrain, Cyprus, Gaza Strip and West Bank, Iran, Iraq, Israel, Jordan, Kuwait, Lebanon, Oman, Qatar, Saudi Arabia, Syrian Arab Republic, United Arab Emirates, Türkiye, and Egypt. Yemen was included in the intended Middle East code list during preprocessing, but it was not present in the final complete-case dataset because of missing required data.

This subgroup was used for visual highlighting and comparison. The study remains global, while Middle Eastern countries are highlighted to examine whether they show a recognizable profile within the global colorectal cancer, lifestyle, and economic space.

## 2.6 Inclusion Criteria and Derived Variables

Countries were included if they had complete observations for country name, ISO3 code, colorectal cancer ASR, adult obesity prevalence, adult physical activity prevalence, GDP per capita, log GDP per capita, and region label. Countries were excluded if any of these core analytical variables were missing. This ensured that the same country set was used across descriptive statistics, visualizations, correlation analysis, regression modeling, and PCA.

Several derived variables were created during processing:

- `cancer_index`: duplicate analytical outcome equal to colorectal cancer ASR;
- `log_gdp`: natural logarithm of GDP per capita;
- `region`: final analytical region label, with the Middle East treated as a separate subgroup;
- `z_cancer_index`, `z_obesity`, `z_physical_activity`, and `z_log_gdp`: standardized variables used for PCA and standardized profile visualization.

Original-scale variables were retained for descriptive statistics, correlation testing, regression modeling, and substantive interpretation. The main data limitations, including cross-sectional design, ecological fallacy, time-lag effects, registry quality, and complete-case filtering, are discussed in the Discussion section.

### 3 Exploratory Data Analysis

#### 3.1 Dataset Overview

The final analytical dataset contained 173 countries after complete-case filtering. The initial merged dataset contained 186 country or territory rows, and 13 rows were excluded because at least one core analytical variable was missing. The final dataset covered six analytical regions: Africa, Americas, Asia, Europe, Middle East, and Oceania. The Middle East subgroup contained 16 countries.

| Section              | Item                       | Value                                                |
|----------------------|----------------------------|------------------------------------------------------|
| Dataset size         | Before filtering           | 186                                                  |
| Dataset size         | After filtering            | 173                                                  |
| Dataset size         | Dropped rows               | 13                                                   |
| Regional structure   | Number of regions          | 6                                                    |
| Regional structure   | Regions                    | Africa, Americas, Asia, Europe, Middle East, Oceania |
| Middle East subgroup | Countries in final dataset | 16                                                   |
| Main variables       | Outcome                    | Colorectal cancer ASR                                |
| Main variables       | Predictors                 | Obesity, physical activity, log GDP                  |

*Table 1: Dataset summary.*

#### 3.2 Descriptive Statistics

*Table 2* shows the basic descriptive statistics for the main continuous variables. Colorectal cancer ASR had a mean of 49.21 and a median of 42.60 per 100,000, showing some right-skewness because a few countries had very high values. Obesity had a mean of 21.83%, while physical activity had a mean of 72.87%. GDP per capita was highly skewed, with the mean much higher than the median, so log GDP was used for the main analysis.

| Variable                     | Mean     | Median  | SD       | Min    | Max       | IQR      |
|------------------------------|----------|---------|----------|--------|-----------|----------|
| Colorectal cancer ASR        | 49.21    | 42.60   | 23.72    | 9.50   | 166.70    | 35.40    |
| Obesity prevalence           | 21.83    | 21.78   | 10.89    | 2.02   | 62.43     | 17.08    |
| Physical activity prevalence | 72.87    | 75.72   | 13.09    | 33.86  | 97.30     | 19.64    |
| GDP per capita               | 16674.82 | 6515.59 | 23570.37 | 302.99 | 123719.66 | 18485.24 |
| Log GDP per capita           | 8.80     | 8.78    | 1.44     | 5.71   | 11.73     | 2.19     |

Table 2: Descriptive statistics.

### 3.3 Regional Distributions

The regional boxplots show how colorectal cancer incidence, obesity, physical activity, and log GDP differ across regions. These figures are mainly descriptive, so the purpose here is to identify broad patterns before formal testing in Section 4.

Europe shows the highest colorectal cancer ASR distribution, while Africa shows the lowest. Asia contains a very high outlier, which later appears clearly in the country-level ranking.

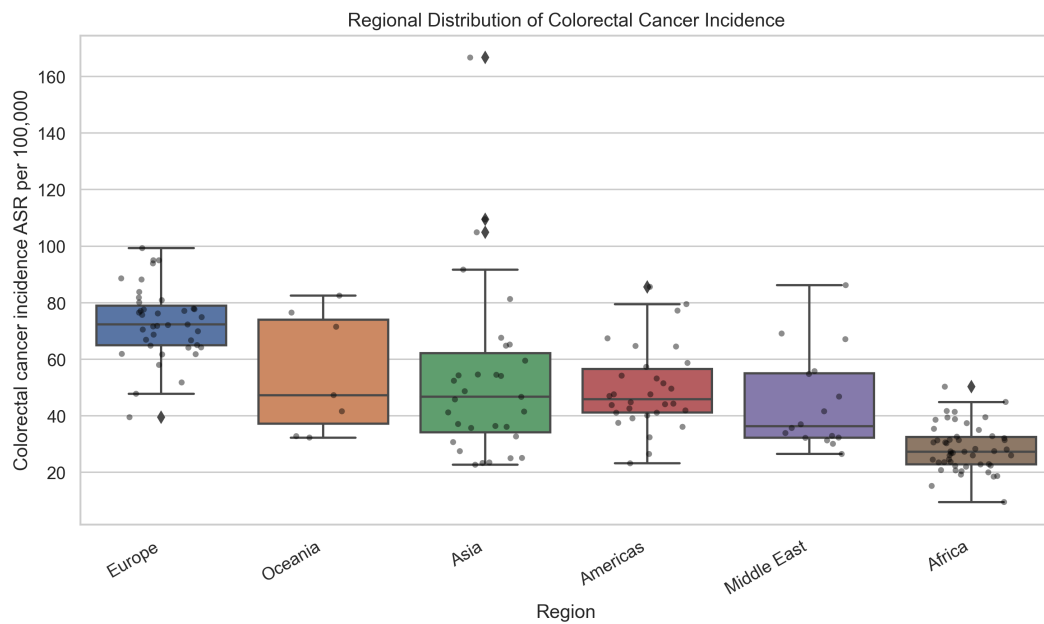
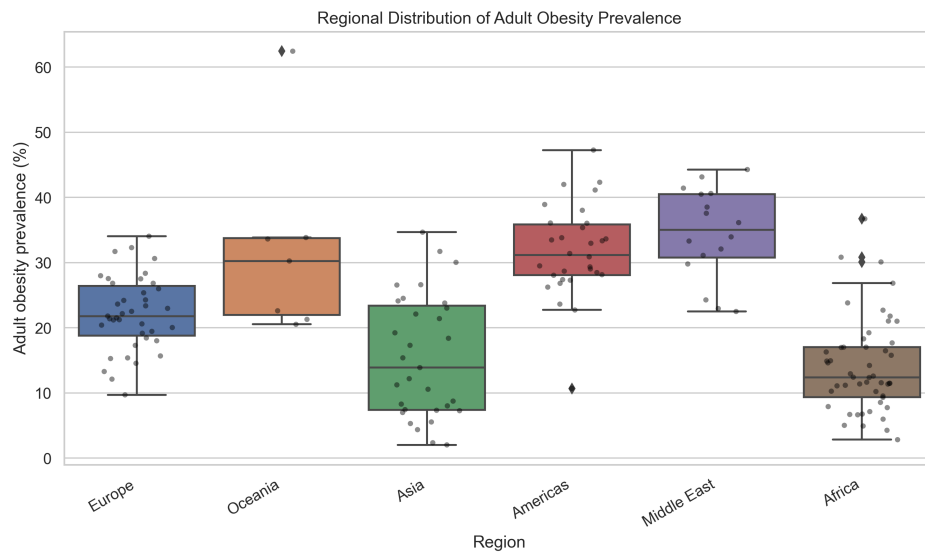


Figure 1: Regional colorectal cancer ASR.

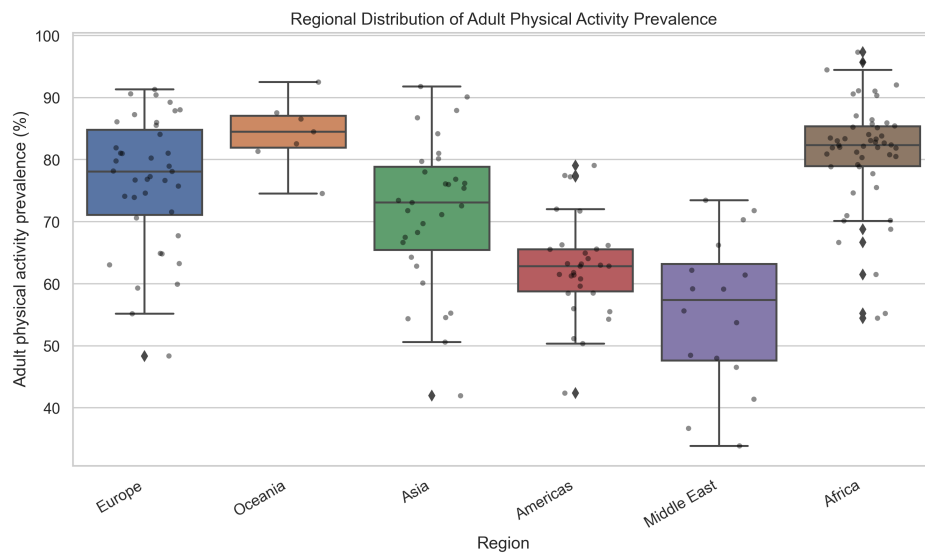
The obesity boxplot shows that the Middle East has one of the highest obesity distributions, while Africa has the lowest. This is important because the Middle East has a high obesity profile, even though its cancer ASR is not the highest globally.





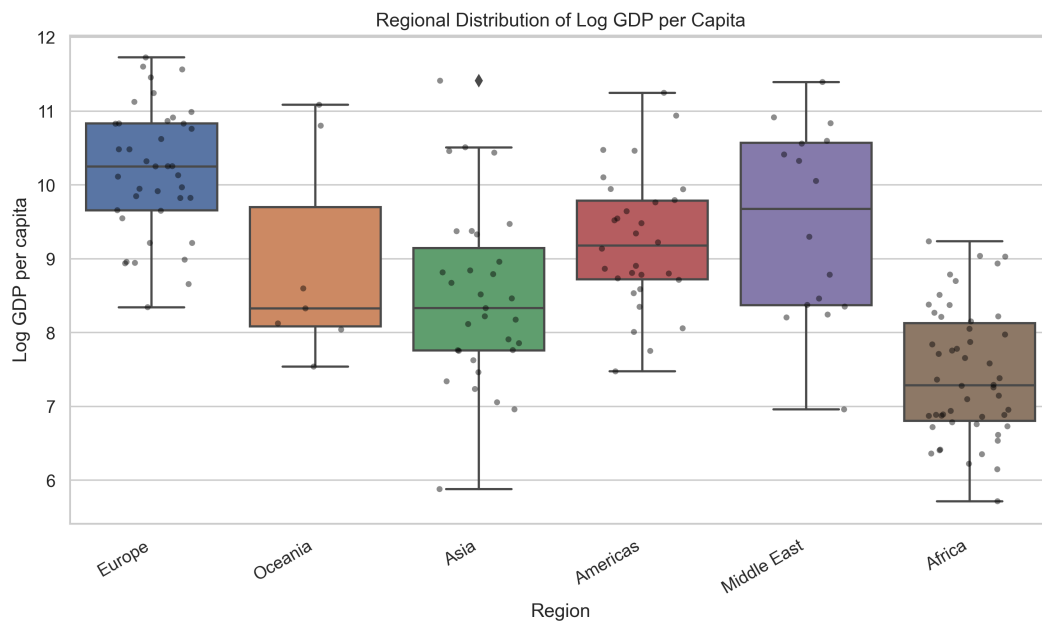
*Figure 2: Regional obesity prevalence.*

The physical activity boxplot shows that the Middle East has one of the lower physical activity distributions. This supports the idea that the region has a distinct lifestyle profile compared with some other regions.



*Figure 3: Regional physical activity prevalence.*

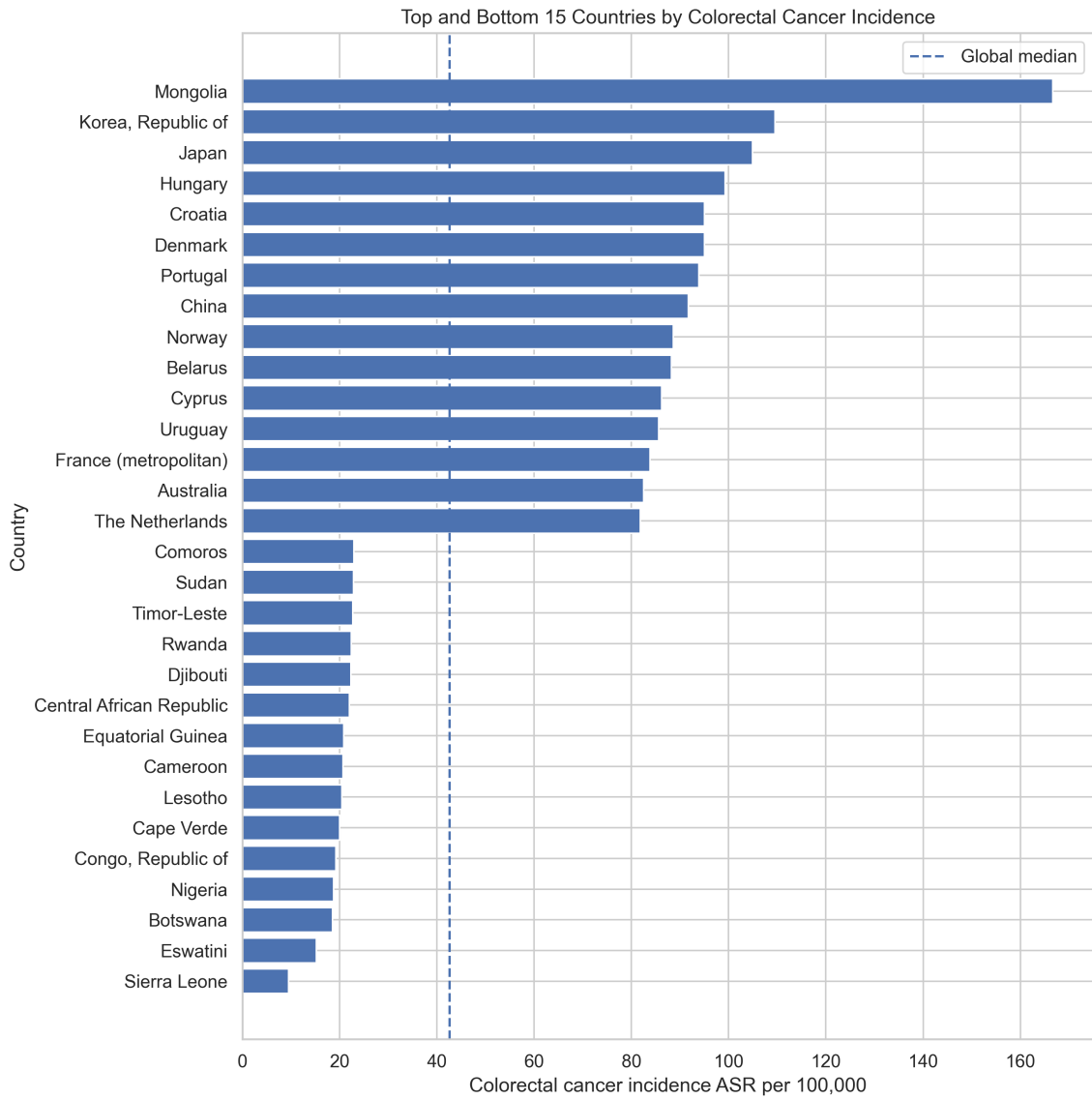
The log GDP boxplot shows that Europe has the highest economic profile, while Africa has the lowest. The Middle East has a wide spread, which reflects the difference between high-income Gulf countries and other Middle Eastern countries.



*Figure 4: Regional log GDP per capita.*

### 3.4 Country-Level Extremes and Middle East Position

The top and bottom country ranking shows the countries with the highest and lowest colorectal cancer ASR. Mongolia is the highest value in the dataset, which explains the high outlier seen in Asia. The lower end is mostly made up of African countries. This figure helps show that individual countries can strongly affect the visual pattern, not only regional averages.



*Figure 5: Top and bottom colorectal cancer ASR countries.*

The Middle East highlighted ranking shows that Middle Eastern countries are spread across the global colorectal cancer distribution. Cyprus, Türkiye, and Israel appear above the global median, while several Gulf countries appear below it. This means the Middle East is not simply a high-incidence or low-incidence region.

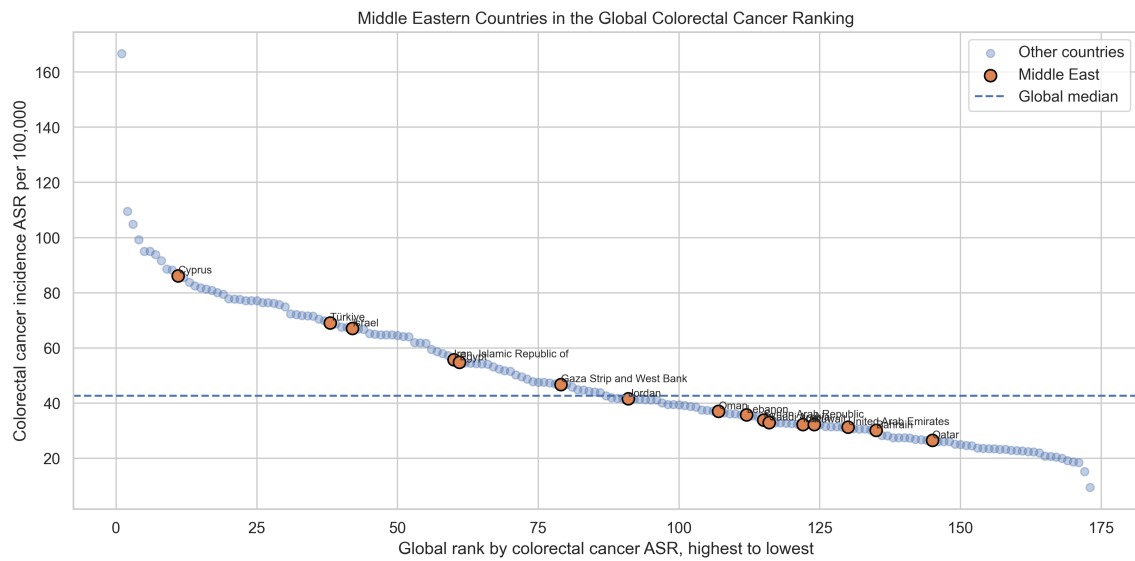


Figure 6: Middle East in global colorectal cancer ranking.

### 3.5 Standardized Regional Cancer and GDP Profile

The standardized regional profile compares colorectal cancer ASR and log GDP on the same scale. Europe is above the global average for both cancer ASR and log GDP, while Africa is below the global average for both. The Middle East has above-average log GDP but slightly below-average cancer ASR, showing that GDP alone does not fully explain regional cancer patterns.

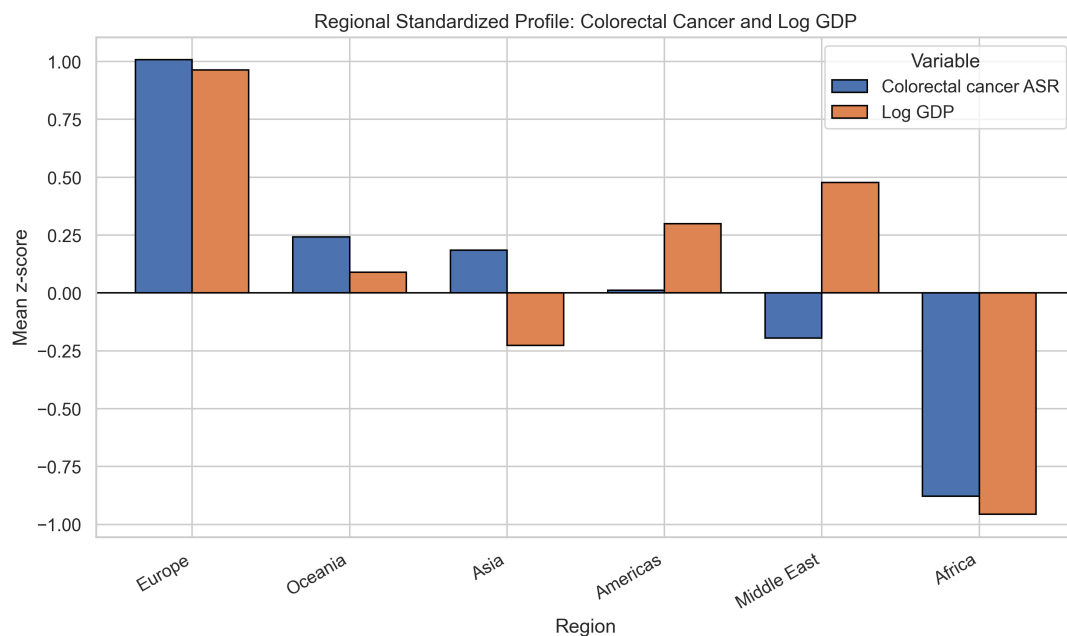


Figure 7: Regional standardized cancer and GDP profile.

### **3.6 EDA Summary**

The exploratory analysis shows that colorectal cancer incidence varies strongly by region, with Europe highest and Africa lowest. The Middle East has high obesity and lower physical activity, but its cancer ASR is mixed rather than uniformly high. Log GDP seems broadly aligned with cancer incidence in some regions, especially Europe and Africa, but the Middle East shows that this pattern is not always direct.

These descriptive findings guide the next stage of analysis. Section 4 uses inferential tests, regression modeling, and PCA to examine these patterns more formally.

## **4 Analysis and Modeling**

This section describes the statistical and modeling methods used after the exploratory data analysis. The aim was to test country-level associations, compare regional differences, and summarize the combined cancer-lifestyle-economic profile of countries. All methods were interpreted as exploratory and ecological, not causal.

### **4.1 Inferential Association Analysis**

Before selecting the main inferential tests, the distributions of colorectal cancer ASR, adult obesity prevalence, adult physical activity prevalence, and log GDP per capita were assessed using the Shapiro-Wilk normality test. This was necessary because the term paper requires test assumptions to be checked rather than assumed. Since global country-level variables can be skewed, especially GDP, the normality check helped justify the use of non-parametric methods.

Spearman rank correlation was used to examine the main bivariate associations. This test was selected because it does not require normal distributions and is suitable for monotonic relationships. The main relationships tested were colorectal cancer ASR with obesity, physical activity, and log GDP. Additional correlations between GDP, obesity, and physical activity were also examined because GDP may reflect both detection capacity and lifestyle transition.

Scatterplots were used alongside the correlation tests to visually inspect these relationships. The plots were colored by region, with Middle Eastern countries highlighted, so that both the global trend and the Middle East position could be examined in the same visual frame. A Spearman correlation heatmap was also used to summarize the direction and strength of the pairwise relationships between the core variables.

Regional differences were tested using the Kruskal-Wallis test. This was chosen because the analysis compared more than two independent regional groups and the variables were not normally distributed. The test was applied to colorectal cancer ASR, obesity prevalence, physical activity prevalence, and log GDP per capita across the six analytical regions. This directly supported the hypothesis that the main cancer, lifestyle, and economic indicators differ by region.

Regression modeling was then used as a multivariable exploratory step. Three models were estimated:

$$\text{Model 1: } cancer\_index = \beta_0 + \beta_1(obesity) + \beta_2(activity) + \epsilon$$

$$\text{Model 2: } cancer\_index = \beta_0 + \beta_1(obesity) + \beta_2(activity) + \beta_3(logGDP) + \epsilon$$

$$\text{Model 3: } cancer\_index = \beta_0 + \beta_1(obesity) + \beta_2(activity) + \beta_3(logGDP) + region + \epsilon$$

Model 1 tested the lifestyle indicators only. Model 2 added log GDP to examine whether the economic indicator improved the explanation of country-level cancer incidence. Model 3 added region and was treated as the main regression model because it included lifestyle, economic, and regional information together. The models were compared using  $R^2$ , adjusted  $R^2$ , and AIC.

## 4.2 Advanced Modeling: Principal Component Analysis

Principal Component Analysis was used as the advanced analytical method required by the assignment guideline. PCA was selected because the study includes several related country-level variables, and the research question asks how countries and regions fit within a combined colorectal cancer, lifestyle, and economic profile space.

The PCA used four standardized variables:

- standardized colorectal cancer incidence ASR;
- standardized adult obesity prevalence;
- standardized adult physical activity prevalence;
- standardized log GDP per capita.

Standardization was required because the variables were measured on different scales. Raw GDP, crude cancer rates, case counts, uncertainty intervals, and region dummy variables were not included in the PCA. This was done so that PCA summarized the core analytical profile rather than population size, uncertainty information, or predefined region labels.

The PCA was evaluated using explained variance, a scree plot, and component loadings. Explained variance showed how much information was captured by each component, while loadings were used to interpret which variables contributed most to each component. The first two components were then used to create a country map. Regional centroids were calculated from average PC1 and PC2 scores to simplify the regional interpretation and support the Middle East profile analysis.

### **4.3 Middle East Supporting Comparison**

Because Middle Eastern countries were highlighted within the global dataset, an additional Middle East versus non-Middle East comparison was conducted using Mann-Whitney U tests. This test was selected because the variables were non-normal and the two groups were unequal in size. The comparison was applied to colorectal cancer ASR, obesity prevalence, physical activity prevalence, and log GDP per capita.

A Gulf versus non-Gulf GDP comparison was also included as a supporting visualization. This was added after the GDP-related plots suggested that Gulf countries formed a visible high-GDP subgroup within the Middle East. This was not treated as a main hypothesis test, but as a supporting check for interpreting internal variation inside the Middle East.

### **4.4 Summary of Analytical Strategy**

Overall, the analysis followed the assignment requirements by combining formal inferential testing with an advanced analytical method. Normality tests guided the choice of non-parametric procedures. Spearman correlations, Kruskal-Wallis tests, Mann-Whitney U tests, and regression models were used for inferential analysis. PCA was used as the advanced method to summarize country profiles and examine the global position of Middle Eastern countries. The results are reported in the next section.

## **5 Results**

This section reports the main inferential results from the final complete-case dataset of 173 countries. The analysis focuses on colorectal cancer incidence, obesity prevalence, physical activity, log GDP per capita, and the Middle East subgroup. The aim is not to prove causation, but to describe whether these country-level variables appear to move together in a meaningful way.

### **5.1 Normality and Choice of Statistical Tests**

The Shapiro-Wilk tests showed that all four main continuous variables were not normally distributed. This supports the use of non-parametric tests, especially Spearman correlation for association and Kruskal-Wallis or Mann-Whitney tests for group comparisons.

| Variable                     | n   | W     | Result     |
|------------------------------|-----|-------|------------|
| Colorectal cancer ASR        | 173 | 0.920 | Non-normal |
| Obesity prevalence           | 173 | 0.978 | Non-normal |
| Physical activity prevalence | 173 | 0.965 | Non-normal |
| Log GDP per capita           | 173 | 0.980 | Non-normal |

Table 3: Shapiro-Wilk normality tests.

## 5.2 Correlation Between Cancer, Lifestyle, and GDP

The Spearman correlation matrix shows that log GDP per capita had the strongest positive relationship with colorectal cancer ASR ( $r_s = 0.677$ ). Obesity also had a positive relationship with colorectal cancer ASR, but it was much weaker ( $r_s = 0.281$ ). Physical activity showed a weak negative relationship with colorectal cancer ASR ( $r_s = -0.177$ ).

This supports the idea that colorectal cancer rates are more strongly linked with economic development than with obesity alone in this dataset. However, this does not mean GDP directly causes cancer. It may reflect better detection, screening, registry quality, and also lifestyle transition in richer countries.

| Variable   | Cancer ASR | Obesity | Activity | Log GDP |
|------------|------------|---------|----------|---------|
| Cancer ASR | 1.000      | 0.281   | -0.177   | 0.677   |
| Obesity    | 0.281      | 1.000   | -0.553   | 0.468   |
| Activity   | -0.177     | -0.553  | 1.000    | -0.361  |
| Log GDP    | 0.677      | 0.468   | -0.361   | 1.000   |

Table 4: Spearman correlation matrix.

The heatmap shows the same correlation results in visual form. The strongest relationship is between log GDP and colorectal cancer ASR, while obesity is positively related to both GDP and cancer ASR.



### Spearman Correlation Matrix of Core Variables Colorectal Cancer Incidence, Obesity, Physical Activity, and Economic Development

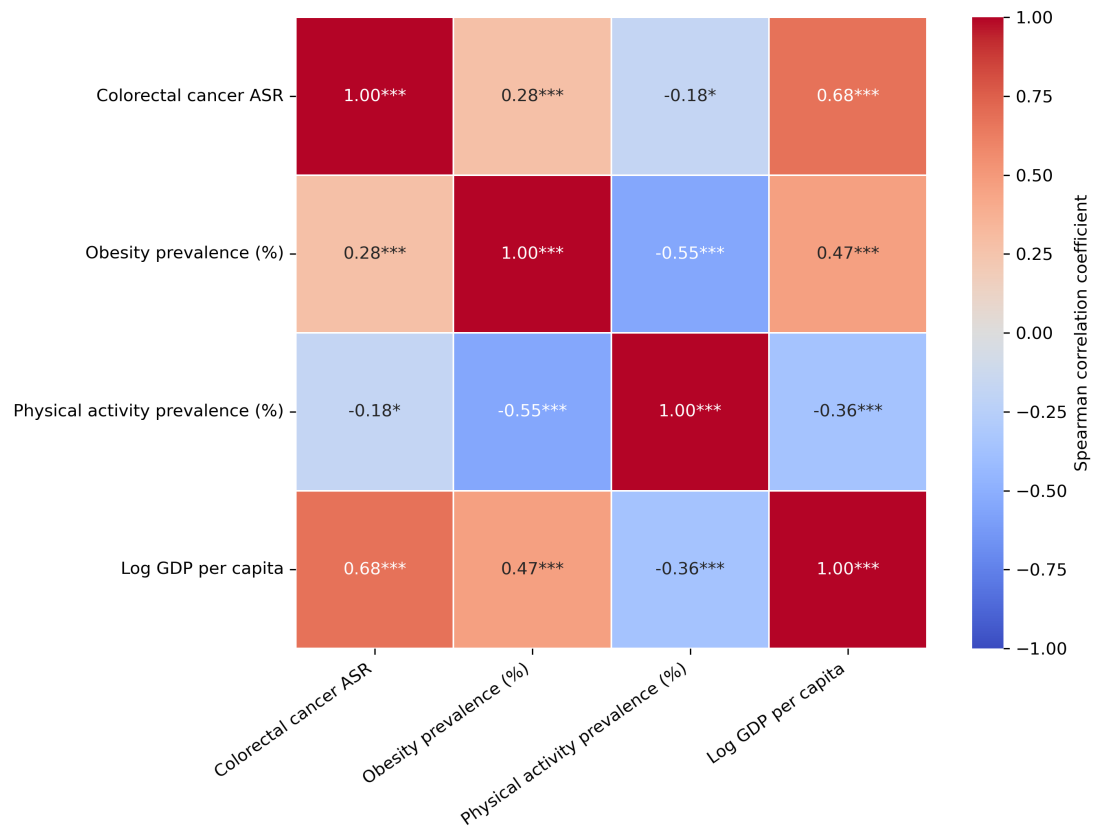


Figure 8: Spearman correlation heatmap.

The obesity plot shows a weak positive trend with colorectal cancer ASR. The relationship is visible, but it is not very strong because many countries with similar obesity levels have different cancer rates.

Figure 5.1: Obesity Prevalence vs. Age-Standardised Cancer Rate

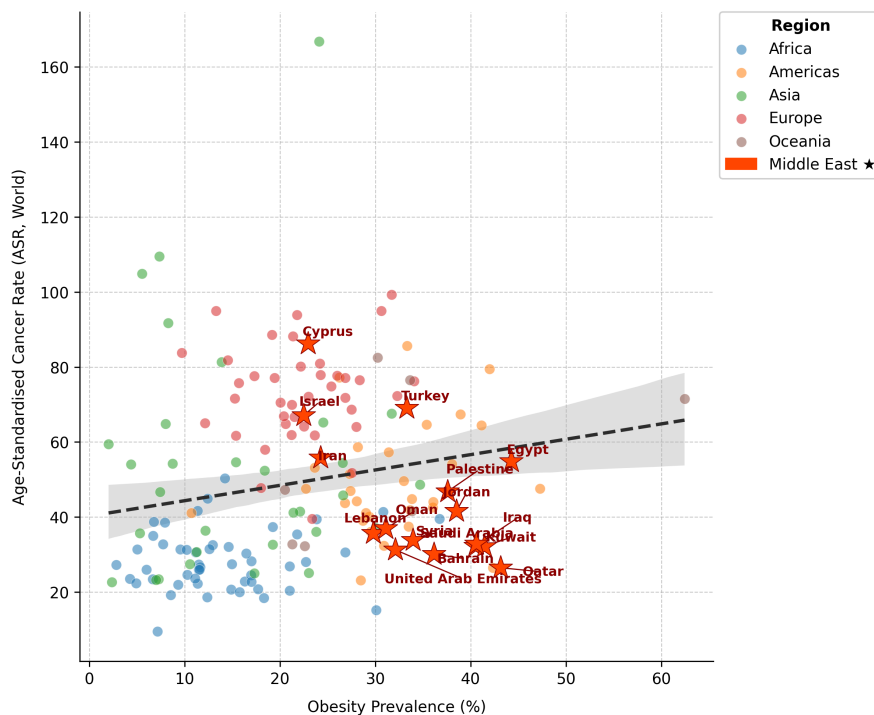


Figure 9: Obesity and colorectal cancer ASR.

The physical activity plot shows a slight downward trend. Countries with higher physical activity often have lower cancer ASR, but the pattern is still weak and scattered.

Figure 5.2: Physical Activity vs. Age-Standardised Cancer Rate

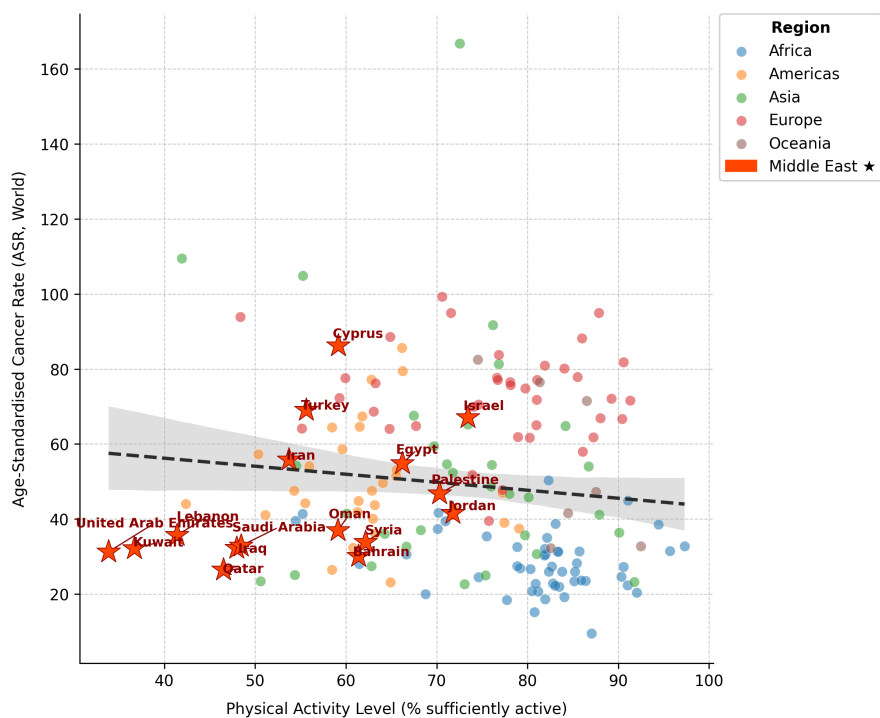


Figure 10: Physical activity and colorectal cancer ASR.

The GDP plot shows the clearest visual pattern. Countries with higher log GDP generally have higher colorectal cancer ASR, which supports the correlation result.

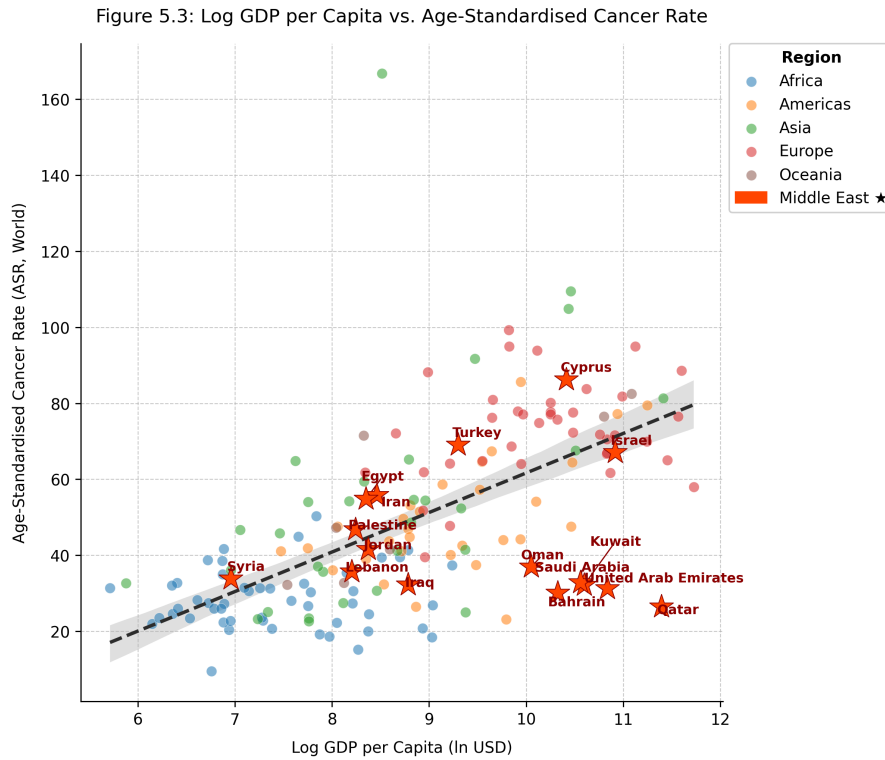


Figure 11: Log GDP and colorectal cancer ASR.

### 5.3 Regional Differences

The Kruskal-Wallis tests show significant regional differences for all four variables. This means colorectal cancer ASR, obesity, physical activity, and log GDP are not evenly distributed across regions. This is expected because regions differ in development level, health systems, lifestyle patterns, and data quality.

| Variable                     | Regions | H      | p-value |
|------------------------------|---------|--------|---------|
| Colorectal cancer ASR        | 6       | 96.388 | < 0.001 |
| Obesity prevalence           | 6       | 90.752 | < 0.001 |
| Physical activity prevalence | 6       | 74.592 | < 0.001 |
| Log GDP per capita           | 6       | 91.563 | < 0.001 |

Table 5: Kruskal-Wallis regional tests.

### 5.4 Regression Results

The regression models show that lifestyle variables alone explained very little of the variation in colorectal cancer ASR. Model 1 had an  $R^2$  of only 0.036. After adding log GDP, the  $R^2$

increased to 0.409. After adding region, the  $R^2$  increased again to 0.533.

This suggests that economic development and regional context explain much more of the country-level cancer differences than obesity and physical activity alone. In the GDP model, log GDP was strongly positive, while obesity became weaker after GDP was included. This may mean that obesity partly overlaps with development level.

| Model                    | n   | $R^2$ | Adj. $R^2$ | AIC     |
|--------------------------|-----|-------|------------|---------|
| Lifestyle only           | 173 | 0.036 | 0.025      | 1585.13 |
| Lifestyle + GDP          | 173 | 0.409 | 0.398      | 1502.53 |
| Lifestyle + GDP + Region | 173 | 0.533 | 0.510      | 1471.78 |

*Table 6: Regression model fit.*

## 5.5 PCA Results

The PCA was used to summarize the four standardized variables into broader country profiles. PC1 explained 53.09% of the variance, and PC2 explained 26.78%. Together, the first two components explained 79.87% of the total variation, which is enough to use the first two components for a simple visual interpretation.

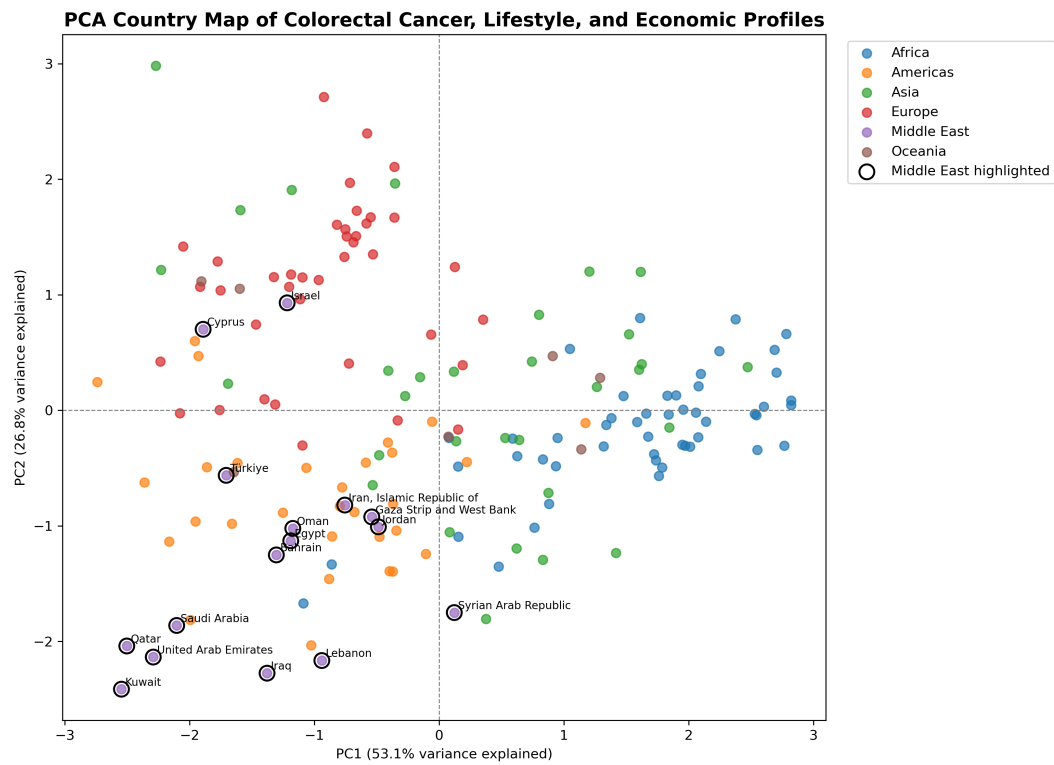
| Component | Variance Ratio | Variance (%) | Cumulative (%) |
|-----------|----------------|--------------|----------------|
| PC1       | 0.531          | 53.090       | 53.090         |
| PC2       | 0.268          | 26.782       | 79.872         |
| PC3       | 0.123          | 12.309       | 92.181         |
| PC4       | 0.078          | 7.819        | 100.000        |

*Table 7: PCA explained variance.*

| Variable                     | PC1    | PC2    | PC3    | PC4    |
|------------------------------|--------|--------|--------|--------|
| Colorectal cancer ASR        | -0.457 | 0.634  | -0.150 | 0.606  |
| Obesity prevalence           | -0.502 | -0.433 | 0.706  | 0.249  |
| Physical activity prevalence | 0.447  | 0.563  | 0.691  | -0.081 |
| Log GDP per capita           | -0.583 | 0.307  | 0.039  | -0.751 |

*Table 8: PCA loadings.*

The PCA country map shows that regions do not completely separate, but there are visible patterns. European countries tend to appear in a higher cancer and higher development area. African countries are more often located toward the lower development and lower cancer side. The Middle East is spread out, but several Gulf countries appear together in a lower cancer but higher GDP and obesity-related profile.



*Figure 12: PCA country map.*

The regional centroid plot simplifies this pattern by showing average regional positions. The Middle East centroid is positioned separately from Europe and Africa. It is closer to a profile with higher obesity and lower physical activity, but not with the highest cancer burden.

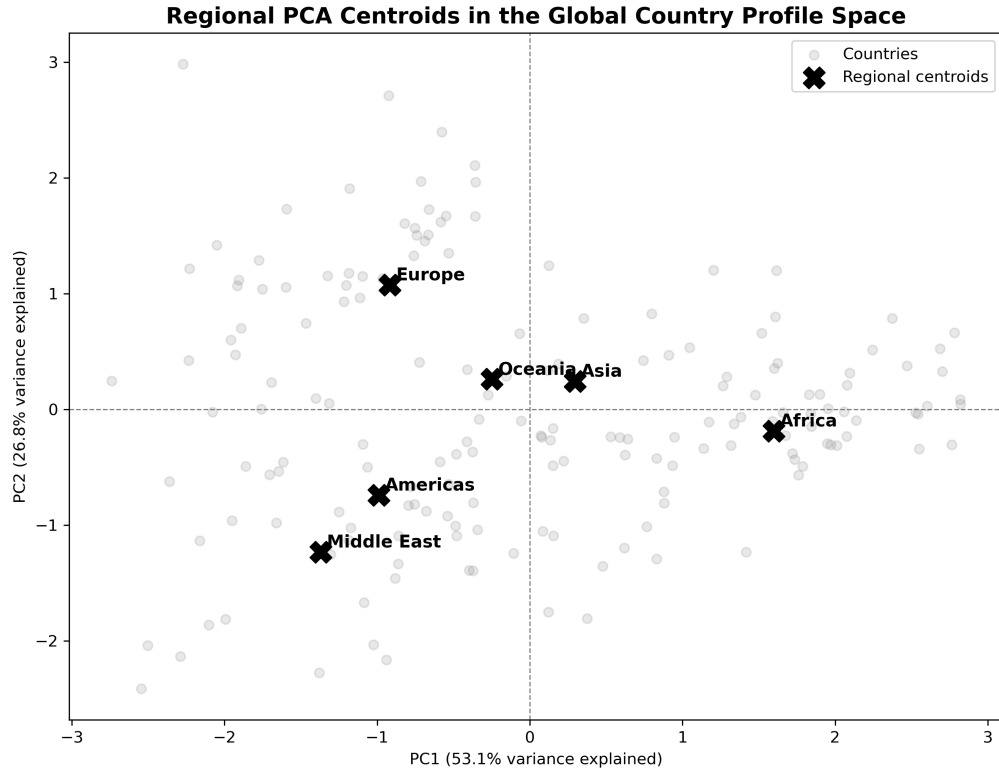


Figure 13: Regional PCA centroids.

## 5.6 Middle East Compared With Non-Middle East

The Middle East subgroup had significantly higher obesity, significantly lower physical activity, and higher log GDP compared with non-Middle Eastern countries. However, colorectal cancer ASR was not significantly different. This is one of the more interesting findings because it shows that high obesity in the Middle East does not automatically translate into higher colorectal cancer ASR in the same cross-sectional year.

| Variable                     | ME median | Non-ME median | ME mean | Non-ME mean | p-value |
|------------------------------|-----------|---------------|---------|-------------|---------|
| Colorectal cancer ASR        | 36.350    | 44.100        | 44.581  | 49.684      | 0.622   |
| Obesity prevalence           | 35.035    | 21.000        | 34.501  | 20.540      | < 0.001 |
| Physical activity prevalence | 57.365    | 77.220        | 55.485  | 74.643      | < 0.001 |
| Log GDP per capita           | 9.675     | 8.713         | 9.485   | 8.733       | 0.048   |

Table 9: Middle East and non-Middle East comparison.

A possible explanation is that some Gulf countries combine high GDP and high obesity with relatively lower recorded colorectal cancer ASR. Higher GDP may be linked to better health-care access and detection systems, but it can also be linked to lifestyle factors such as richer diets, sedentary work, and more calorie-dense food environments. Therefore, GDP seems to work in two directions in this paper: it may reflect both detection capacity and lifestyle transi-

tion.

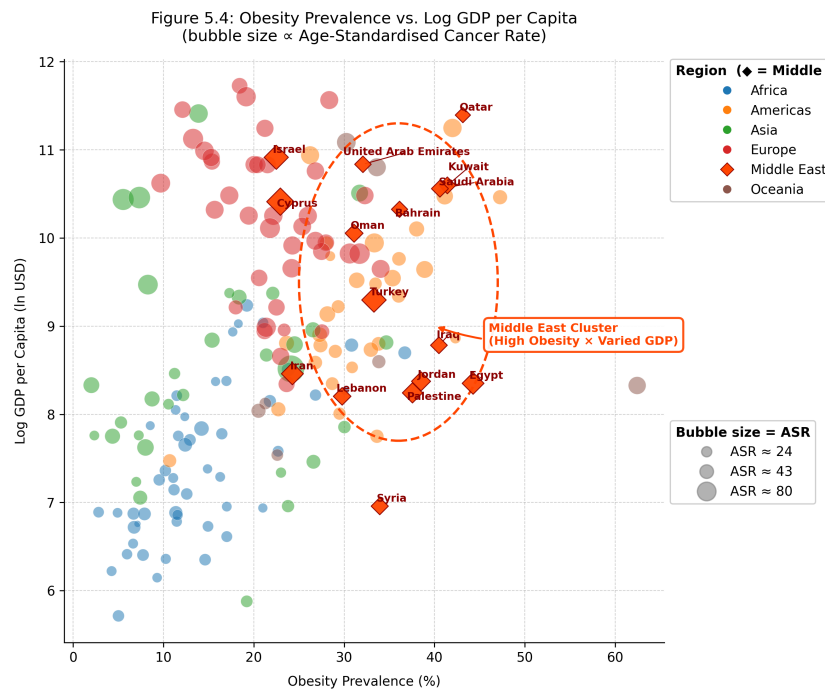


Figure 14: Obesity, log GDP, and cancer ASR overlap.

The Gulf and non-Gulf comparison was added after the plots showed that several Gulf countries seemed to form a separate cluster. The GDP comparison supports this observation: Gulf countries have much higher GDP per capita than the other selected Middle Eastern countries. This helps explain why Gulf countries may behave like a special subgroup inside the Middle East rather than following exactly the same pattern as the rest of the region.

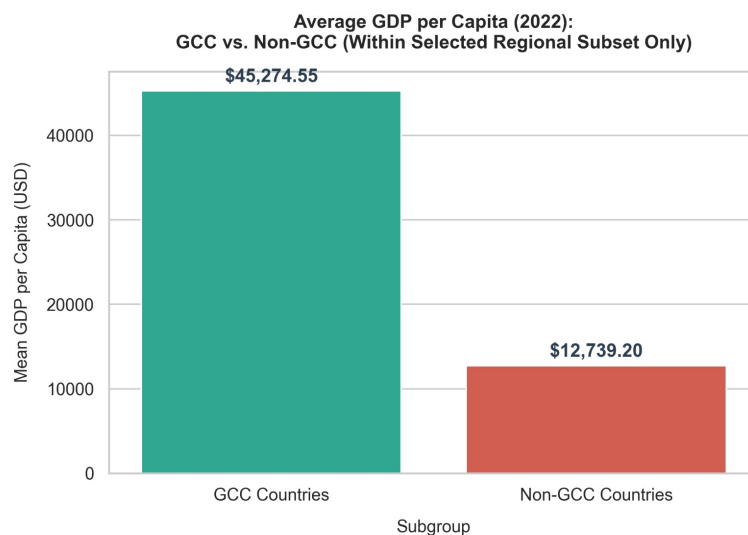


Figure 15: GCC and non-GCC GDP comparison.

Overall, the results show that colorectal cancer incidence is most strongly related to GDP and region in this global dataset. Obesity is still relevant, but its direct relationship with cancer

is weaker after considering GDP. For the Middle East, the main pattern is high obesity, lower physical activity, and relatively high GDP, but not clearly higher colorectal cancer ASR. This supports the idea that the relationship between obesity and colorectal cancer is influenced by wider economic and regional factors.

## **6 Discussion**

This study explored global country-level patterns in colorectal cancer incidence using obesity prevalence, physical activity prevalence, GDP per capita, and regional grouping. The main result was that log GDP per capita had the strongest and most consistent association with colorectal cancer ASR. Obesity showed a weaker positive association, while physical activity showed a weak negative association. Regional differences were also clear, and the Middle East appeared to have a somewhat distinct profile because it had high obesity and lower physical activity, but not clearly higher colorectal cancer ASR compared with the rest of the world.

Overall, these results fit the idea that colorectal cancer incidence is not explained by one factor only. A simple explanation like “higher obesity means higher cancer” is not enough for this dataset. Instead, the patterns seem to involve a mixture of lifestyle, economic development, healthcare detection, and regional context.

### **6.1 Interpretation of Main Findings**

The strongest pattern in the analysis was the positive relationship between log GDP per capita and colorectal cancer ASR. This was visible in the scatterplot, the Spearman correlation matrix, and the regression models. This does not mean that GDP directly causes colorectal cancer. A more careful interpretation is that GDP works as a country-level proxy for several things at the same time.

The first pathway is detection capacity. Countries with higher GDP often have stronger health-care systems, better access to diagnosis, more screening, and more complete cancer registries. Therefore, they may detect and report more colorectal cancer cases. This may partly explain why high-income countries tend to show higher recorded colorectal cancer incidence. This interpretation is also consistent with previous global cancer studies and GLOBOCAN-based analyses that show large international differences in cancer incidence and detection capacity (Arnold et al., 2017; Bray et al., 2024; Center et al., 2009).

The second pathway is lifestyle transition. Higher GDP may also reflect changes in diet, work, transport, and daily activity. Richer or more urbanized countries may have more sedentary lifestyles and more access to calorie-dense, processed, or fast-food diets. This could help explain why GDP was also positively related to obesity in this study. Previous research has also connected colorectal cancer risk with body fatness, low physical activity, and diet-related factors (Larsson & Wolk, 2007; Moghaddam et al., 2007; Shaw et al., 2018; World Cancer Research Fund/American Institute for Cancer Research, 2018).



The obesity result was in the expected direction, but it was not as strong as GDP. Obesity had a positive relationship with colorectal cancer ASR in the correlation analysis, but it became weaker after GDP was added to the regression model. This suggests that obesity and GDP overlap in this dataset. In other words, richer countries may often have both higher obesity and better cancer detection, so it becomes difficult to separate these effects using only one year of country-level data.

Physical activity had a weak negative relationship with colorectal cancer ASR. This direction was expected because physical activity is usually considered protective against colorectal cancer (Shaw et al., 2018; World Cancer Research Fund/American Institute for Cancer Research, 2018). However, the relationship was not very strong in the country-level data. One possible reason is that the physical activity variable is measured at the population level and may not directly capture the long-term exposure patterns that matter for cancer development.

## **6.2 Middle East Interpretation**

The Middle East result was one of the most interesting parts of the analysis. The region had significantly higher obesity and lower physical activity compared with non-Middle Eastern countries, but its colorectal cancer ASR was not significantly higher. This means that the Middle East does not simply follow the expected lifestyle-risk pattern.

This could happen for several reasons. First, colorectal cancer usually develops over a long time, so the current 2022 obesity and physical activity values may not match cancer incidence in the same year. Second, countries may differ in screening, diagnosis, and cancer registration quality. Third, the Middle East itself is not one uniform group. Gulf countries have much higher GDP than many non-Gulf Middle Eastern countries, which may place them differently in the global cancer-lifestyle-economic space.

The Gulf comparison was useful here. It showed that Gulf countries have much higher GDP than non-Gulf Middle Eastern countries. This helps explain why some Gulf countries appeared as a separate subgroup in the GDP-related plots. At the same time, high GDP can mean two different things: better healthcare capacity on one side, and more lifestyle transition on the other side. This makes the Middle East pattern more mixed and more interesting than expected.

## **6.3 Hypothesis Evaluation**

H1 stated that countries with higher adult obesity prevalence would tend to have higher colorectal cancer incidence ASR. This hypothesis was partially supported. Obesity had a positive correlation with colorectal cancer ASR, and the scatterplot also showed a weak positive trend. However, the association was not strong, and it became weaker when GDP was included in the regression model.

H2 stated that countries with lower adult physical activity prevalence may tend to have higher colorectal cancer incidence ASR. This hypothesis was also partially supported. The correlation

between physical activity and colorectal cancer ASR was negative, which is in the expected direction. However, the relationship was weak, so the evidence is not strong.

H3 stated that GDP per capita would be associated with colorectal cancer ASR through detection capacity and lifestyle transition. This hypothesis was strongly supported. GDP had the strongest correlation with colorectal cancer ASR and remained important in the regression models. The results also support the idea that GDP may reflect both better healthcare detection and broader lifestyle changes.

H4 stated that colorectal cancer, obesity, physical activity, and GDP patterns would differ across world regions. This hypothesis was supported. The Kruskal-Wallis tests showed significant regional differences for all four variables. The EDA boxplots and PCA results also showed that regions had different profiles.

H5 stated that Middle Eastern countries would occupy a semi-distinct position in the global colorectal cancer, lifestyle, and economic profile space. This hypothesis was supported, but with caution. The Middle East did not form a completely separate cluster, but it showed a recognizable profile in the PCA and subgroup comparisons. It was especially different because of high obesity, lower physical activity, and relatively high GDP, while colorectal cancer ASR was not significantly higher than the rest of the world.

## **6.4 Comparison With Prior Literature**

The findings are broadly in line with previous global studies showing that colorectal cancer incidence is often higher in more developed countries or countries with higher human development indicators (Arnold et al., 2017; Center et al., 2009; Goodarzi et al., 2019; Khazaei et al., 2016; Rafiemanesh et al., 2016). These studies usually explain the pattern through a mixture of lifestyle differences, health system differences, and better detection.

The present study adds to this by using GDP per capita directly as a simple economic indicator and combining it with obesity, physical activity, regional labels, and a Middle East highlight. The result is not completely new in the sense that development and colorectal cancer have been studied before. However, the contribution here is the combined visual analytics approach and the specific positioning of Middle Eastern countries within the global profile.

The lifestyle findings also fit the general literature. Obesity has been linked with increased colorectal cancer risk in meta-analyses (Larsson & Wolk, 2007; Moghaddam et al., 2007), while physical activity has been associated with lower colorectal cancer risk (Shaw et al., 2018). However, in this paper, lifestyle variables were weaker than GDP. This does not mean lifestyle is unimportant. It mainly shows that country-level, same-year data may not capture the long-term individual exposure patterns that matter for cancer development.

For the Middle East, the results connect with previous work showing that modifiable risk factors such as obesity, physical inactivity, and diet are important concerns in the MENA region (Haghighatdoost et al., 2024; Mansour et al., 2024). At the same time, the present analysis shows that high obesity and lower physical activity do not automatically appear as higher

recorded colorectal cancer ASR in the same cross-sectional year. This is why screening, registry quality, GDP differences, and time-lag effects are important when interpreting the Middle East pattern.

## **6.5 Methodological Interpretation**

A major point in interpreting this study is that the data are ecological. Each row is a country, not a person. Therefore, the results cannot be used to say that an individual with a certain BMI or activity level has a certain cancer risk. This is the ecological fallacy problem. The results only describe patterns between countries.

Another important point is the time lag. Colorectal cancer can develop over many years. Obesity and physical activity in 2022 may not represent the relevant exposure history that contributed to cancer incidence in 2022. A country may have high obesity now, but its cancer incidence may reflect lifestyle patterns from many years earlier.

GDP should also be interpreted carefully. In this paper, GDP is not treated as a biological cause of cancer. It is used as a proxy for wider country-level conditions, including healthcare access, screening, diagnostic capacity, registry quality, and lifestyle transition. Because GDP captures many things at once, it is useful but also difficult to interpret.

## **6.6 Limitations**

This study has several limitations. First, it is cross-sectional, so it only looks at one year. This makes it impossible to study trends or time-lag effects properly. Second, the analysis uses country-level data, so it cannot explain individual-level cancer risk. Third, the dataset only included variables that were available consistently across countries. Many other relevant factors were not included, such as smoking, alcohol consumption, red and processed meat intake, screening coverage, urbanization, healthcare expenditure, education, and genetic or demographic differences.

Another limitation is that cancer incidence data can depend on registry quality. Countries with weaker cancer registration systems may underreport cancer incidence. This means that lower recorded cancer rates do not always mean lower true cancer burden. Finally, the Middle East subgroup had only 16 countries, so subgroup results should be interpreted carefully.

## **6.7 Future Work**

Future research could expand this analysis in several directions. A time-series study would be useful to examine how colorectal cancer incidence, obesity, physical activity, and GDP change over time. This would help address the time-lag issue more directly.

A more focused Middle East study would also be useful. Such a study could include additional regional variables such as diet, screening programs, healthcare expenditure, urbanization, dia-

betes prevalence, and country-specific cancer registry quality. It could also separate Gulf and non-Gulf countries more formally.

Another future direction would be to study a wider group of related cancers, such as gastrointestinal cancers, and compare whether similar GDP and lifestyle patterns appear. This could show whether the pattern found here is specific to colorectal cancer or part of a broader regional cancer profile.

## **7 Conclusion**

This paper examined global country-level patterns in colorectal cancer incidence in 2022 using obesity prevalence, physical activity prevalence, GDP per capita, and regional grouping. The main finding was that log GDP per capita had the strongest association with colorectal cancer ASR. Obesity showed a weaker positive association, while physical activity showed a weak negative association.

The Middle East showed a semi-distinct profile. It had high obesity, lower physical activity, and relatively high GDP, but it did not have significantly higher colorectal cancer ASR compared with the rest of the world. This suggests that colorectal cancer patterns are shaped by more than lifestyle indicators alone. Economic development, healthcare detection capacity, and regional context also matter.

Overall, this study provides a reproducible visual analytics pipeline for exploring colorectal cancer incidence across countries. The findings should be interpreted as ecological associations, not causal conclusions.

## **Reproducibility Statement**

The analysis was conducted using a reproducible data processing and visualization workflow. The project repository is available at:

<https://github.com/raihan-karim-ishmam/global-colorectal-cancer-eda>

The repository contains the code used for data cleaning, variable transformation, statistical testing, regression modeling, PCA, and visualization. It also includes the processed dataset, generated figures, output tables, and supporting files. A README file is provided in the repository to explain the project structure and reproduction steps.

For submission, the paper is provided in both PDF and DOCX formats. The bibliography source file, `references.bib`, is also submitted to support transparent reference management and citation checking.

## AI Use Disclosure

AI-assisted tools were used for grammar refinement, LaTeX coding of text and tables, and code debugging support. AI was also used as a search support tool to identify relevant libraries and literature sources. All substantive methodological decisions, data interpretation, results verification, and final writing responsibility remain with the authors.

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## A Appendix

This appendix includes supporting outputs that were useful for checking and explaining the main analysis, but were not necessary to keep in the main results section.

### A.1 Regression Coefficients

The main Results section reports the model fit comparison. The coefficient table is included here to show the main terms used in the three regression models predicting colorectal cancer incidence ASR.

| Model                    | Term              | Coef.  | p-value |
|--------------------------|-------------------|--------|---------|
| Lifestyle only           | Obesity           | 0.375  | 0.049   |
| Lifestyle only           | Physical activity | -0.056 | 0.722   |
| Lifestyle + GDP          | Obesity           | -0.154 | 0.330   |
| Lifestyle + GDP          | Physical activity | 0.140  | 0.267   |
| Lifestyle + GDP          | Log GDP           | 11.334 | < 0.001 |
| Lifestyle + GDP + Region | Obesity           | -0.051 | 0.774   |
| Lifestyle + GDP + Region | Physical activity | -0.006 | 0.966   |
| Lifestyle + GDP + Region | Log GDP           | 6.342  | < 0.001 |

*Table 10: Regression coefficients for models predicting colorectal cancer incidence ASR.*

### A.2 Geographic View of Colorectal Cancer Incidence

The geographic heat map is included as a supporting visualization. It gives a quick global view of where colorectal cancer incidence ASR is higher or lower, but the main analysis relies on regional boxplots, correlations, regression, and PCA.

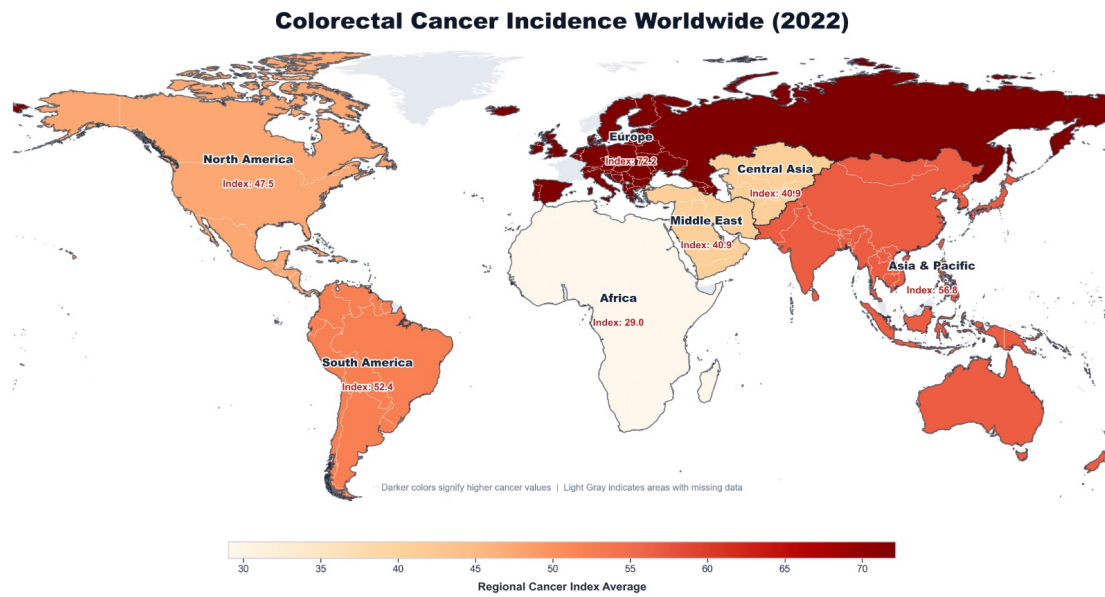


Figure 16: Geographic heat map of colorectal cancer incidence ASR.

### A.3 Gulf and Non-Gulf Middle Eastern Countries

This figure supports the interpretation that the Middle East is not one uniform group. It was included after the GDP-related plots suggested that several Gulf countries formed a visible high-GDP subgroup within the Middle East.

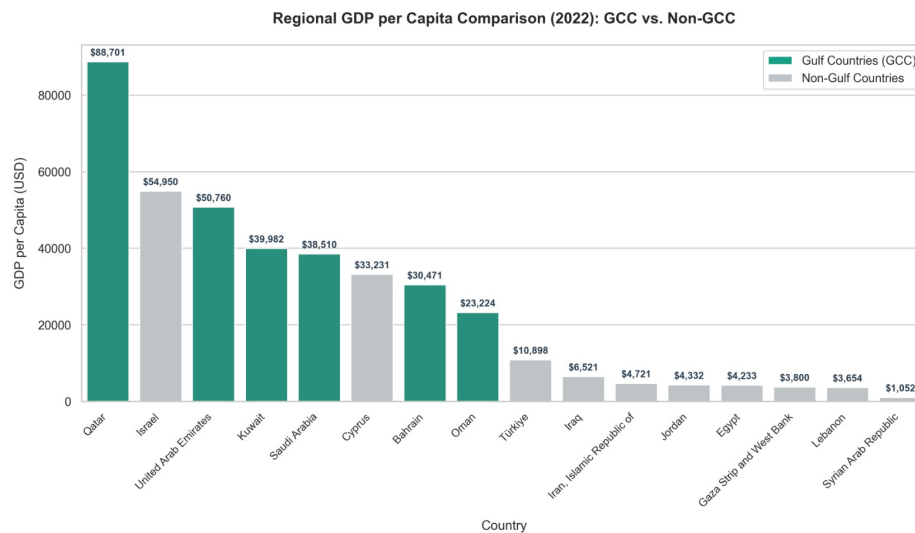


Figure 17: Gulf and non-Gulf Middle Eastern countries.